



More deprived, more billboards?

Mapping Billboard Exposure Inequality in Central London using Street View Images and Deep Learning

Research Problem

What?

Billboards are part of our everyday urban environment (Cronin, 2006).

The exposure of billboards are not evenly distributed (Hasanov et al. 2025).

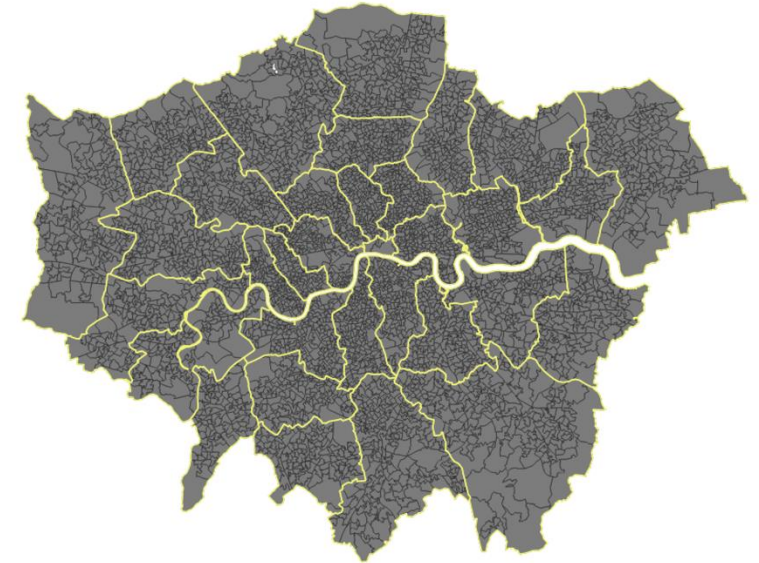
Why it matters?

Unequal exposure may reinforce social inequality (Palmer et al, 2021)

Manual auditing is slow and expensive (Hasanov et al. 2025).

Where?

London is a suitable case study because it combines dense street-view coverage with marked neighbourhood inequality (Crewe and Wang, 2018).



Related works- Method

Hasanov et al. (2025)

Developed an automated system for detecting outdoor advertisement banners from street-level video.

Compared classical computer vision, LLM-based analysis, and a custom-trained YOLOv11 model.

Found that YOLOv11 performed best for banner detection and could be linked to GPS-based geospatial mapping.

Limitation

Small and city-specific training data:

the model is trained on a limited dataset from Baku, which restricts generalisability.

Relatively weak evaluation design:

the study reports limited metrics, so model performance is not assessed as rigorously as in standard detection research.

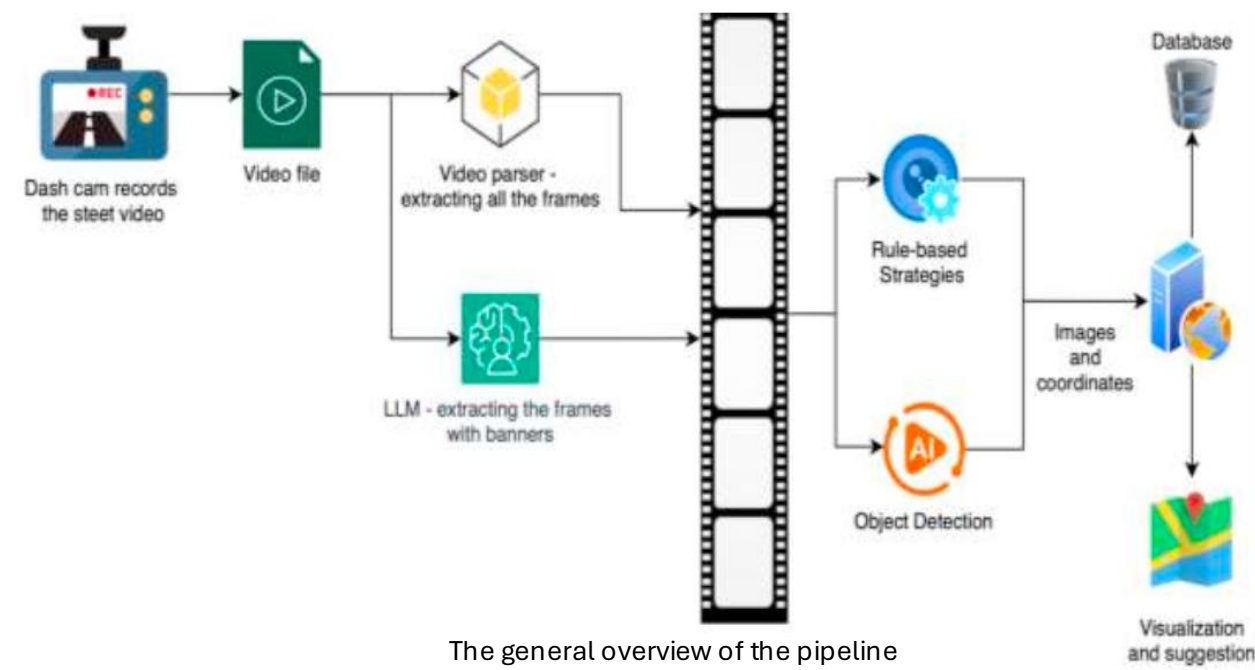


Table 2. Comparison of the functional performance of different approaches

Approach	Accuracy	Precision
Classic Computer Vision Algorithms	50%	73%
LLM (GPT-4)	64%	68%
LLM (Llama)	61%	66%
YOLOv11	92%	89%

Why it matters for our project ?

Outdoor advertisement monitoring can be automated.

Specialised vision models are more practical than LLM-based approaches for this task.

Related works- Liverpool

Palmer et al. (2021)

Used street-view images and deep learning to identify and classify outdoor advertisements in Liverpool.

Linked advertisement patterns to neighbourhood deprivation.

More advertisements in deprived areas.

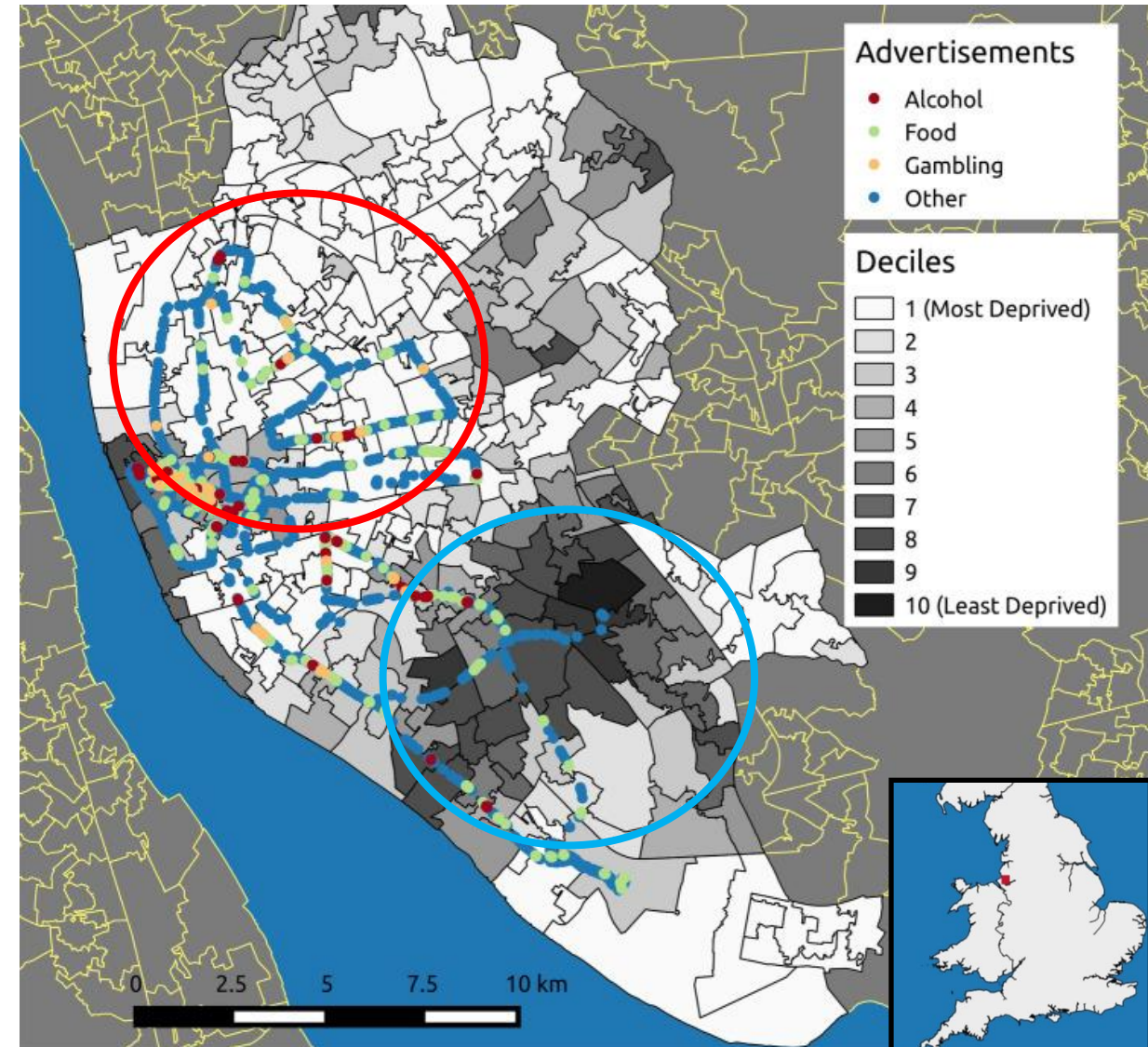
Limitation

The data collection is not fully representative: the Liverpool street-view dataset is skewed towards more deprived areas,

Why it matters for our project?

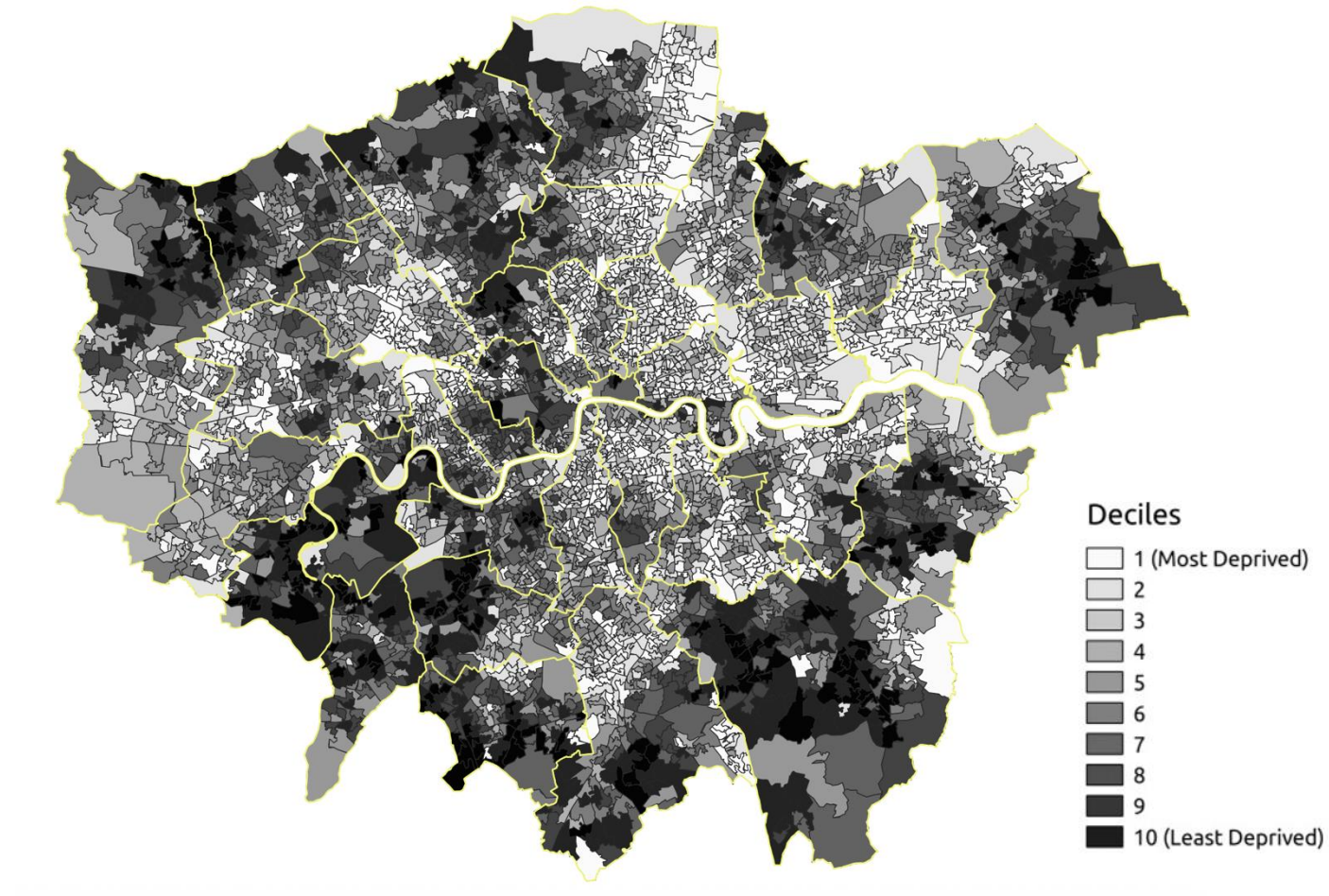
Billboard exposure can be studied at scale using street-view imagery.

Advertising exposure may be socially uneven.



Liverpool advertisement locations by Lower Super Output Areas (LSOAs). A color gradient indicates the level of deprivation, with white and black being the most and least deprived respectively.

Liverpool showed a pattern — but what about a larger, more complex city like London?



IMD Deciles in London

Our Project- More deprived, more billboards?

Study area: Central London

Research focus: billboard exposure
inequality

Approach: a sampled assessment
using street-view imagery



IMD Deciles in Central London

Task definition

1. Task type

Binary image classification

2. Input

Street-view images sampled across Central London

3. Output / labels

1 = billboard present

0 = no billboard

4. Ground truth

Manually annotated with binary labels to serve as training and validation data.

5. Prediction

The trained models predict on the remaining unlabelled images

Dataset

Source:

1. ~2000 points (0°, 90°, 180°, 270°)= Street-level imagery via the Google Street View API (~8000)
2. Road network data from Transport for London (TLRN)
3. Sampling points generated along major roads across Central London
4. Index of Multiple Deprivation (IMD) data for neighbourhood deprivation measurement

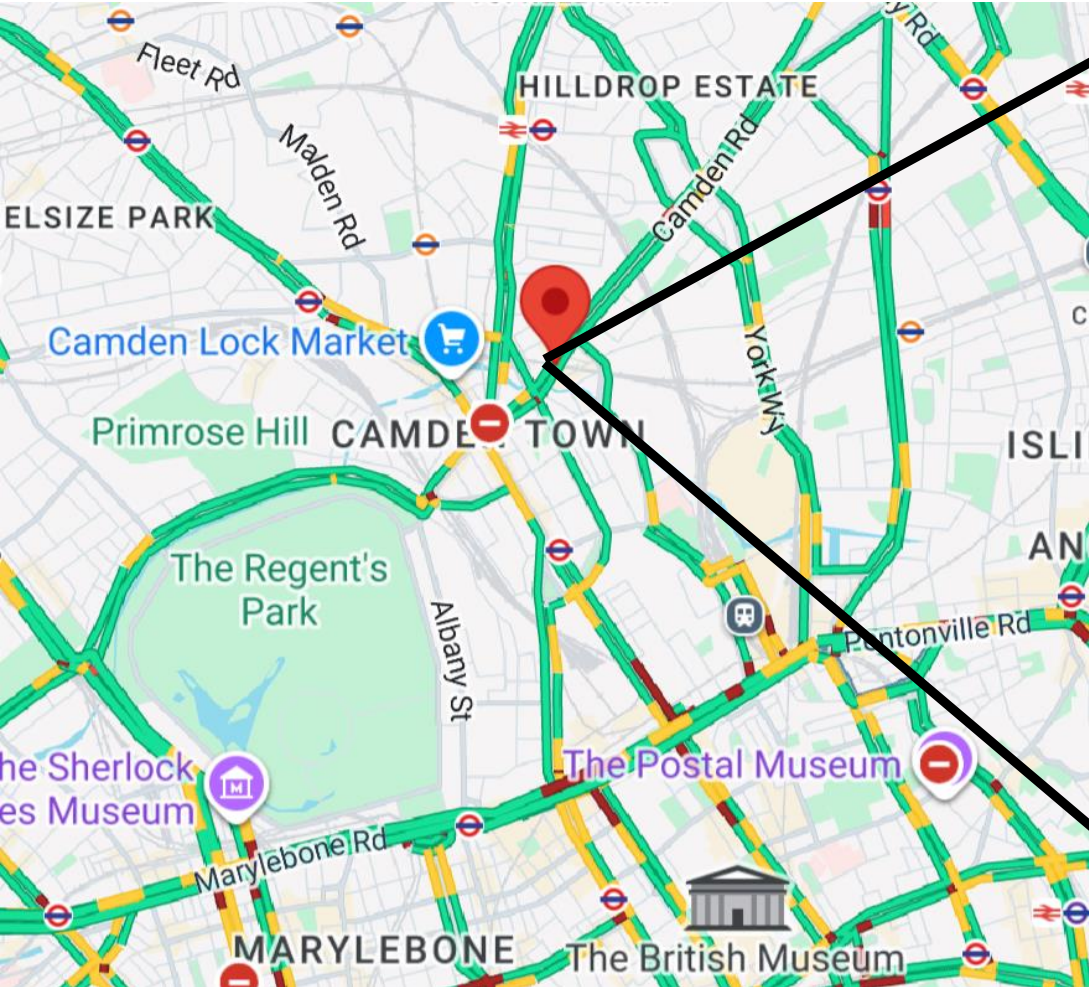
Manual Annotation:

1. ~375 points (0°, 90°, 180°, 270°) = ~1500 manually labelled as ground truth,
2. Binary labels: billboard present (1) or absent (0)

Ethics/licensing:

1. Images are sourced from publicly accessible street-view services subject to platform licensing terms
2. The analysis focuses on public-facing billboard exposure rather than individuals

Dataset- example



A point in Camden Town (51.541944 -0.138503)

View Angle: 0°

Billboard present

View Angle: 90°

View Angle: 180°

View Angle: 270°

Billboard present

Four images downloaded by Street View Static API

Model approach

Data preprocessing

Resizing & Normalization: Images resized to 224×224; normalised to ImageNet statistics.

Train / Test split: 80% / 20% (with stratify due to sparsity of billboards)

1. Model Selection

Task: binary image classification

Model 1: ResNet-18 — a strong and efficient CNN baseline (He et al., 2016)

Model 2: VGG16 — a classic deep CNN for comparison (Simonyan and Zisserman, 2015)

Model 3: ViT — a transformer-based model to test whether a non-CNN architecture offers added value (Dosovitskiy et al., 2020)

2. Output

The model predicts the probability that a billboard is present in each image

3. Decision rule

probability $\geq 0.5 \rightarrow$ billboard present

probability $< 0.5 \rightarrow$ no billboard

4. Evaluation

Accuracy

Precision

Recall

F1-score

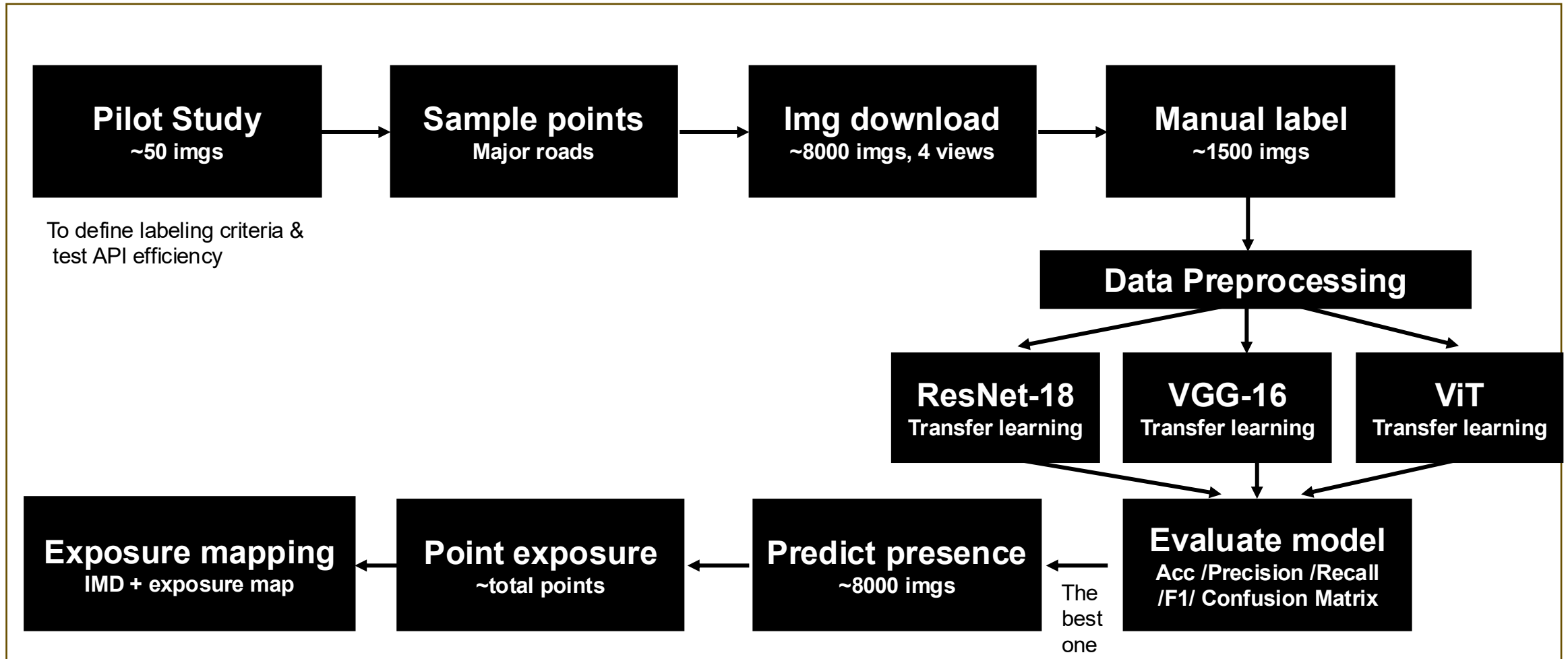
Confusion Matrix

Point exposure

Predictions from four views are averaged to estimate point-level exposure

Views: 0°, 90°, 180°, 270°

Pipeline



Reference

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- Cronin, A. M. (2006) Advertising and the Metabolism of the City: Urban Space, Commodity Rhythms. *Environment and Planning D: Society and Space* 24(4): 615–632.
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- Palmer, G., Green, M., Boyland, E., Vasconcelos, Y. S. R., Savani, R., and Singleton, A. (2021) A deep learning approach to identify unhealthy advertisements in street view images. *Scientific Reports* 11(1): 4884.
- Simonyan, K., and Zisserman, A. (2015) Very deep convolutional networks for large-scale image recognition. In *International Conference on Learning Representations (ICLR 2015)*.